PART 3 Instructions

How do you earn points?

There are two factors that determine your points:

(1) How many Green balls are drawn:

Once you submit your decision on how many balls to draw from each urn, the computer draws 100 balls accordingly. If a drawn ball is green, the green ball awards you a positive number of points. If a drawn ball is black, the Black ball awards you 0 points.

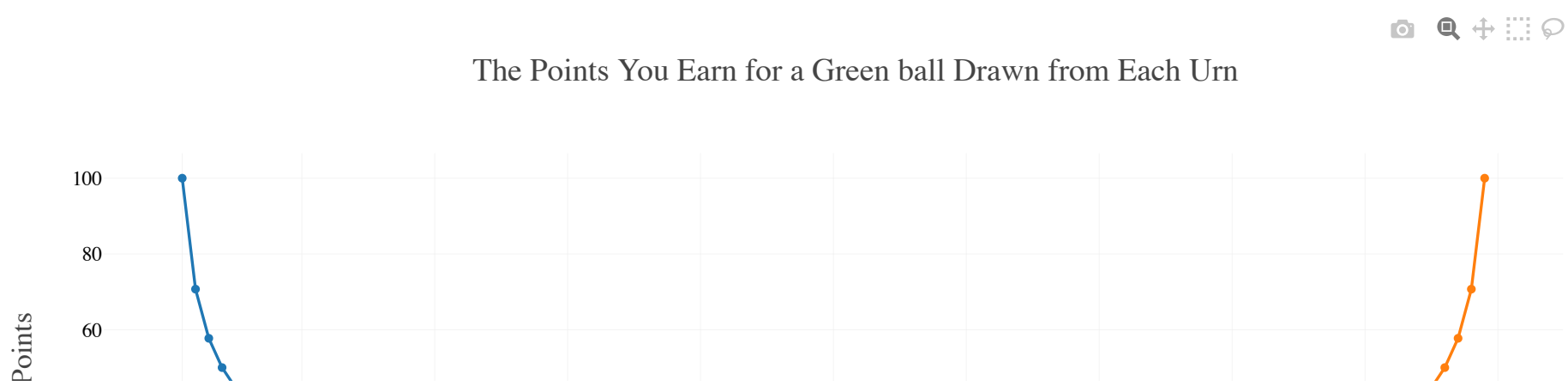
*Therefore, you will earn more points as you get to draw more **Green** balls.*

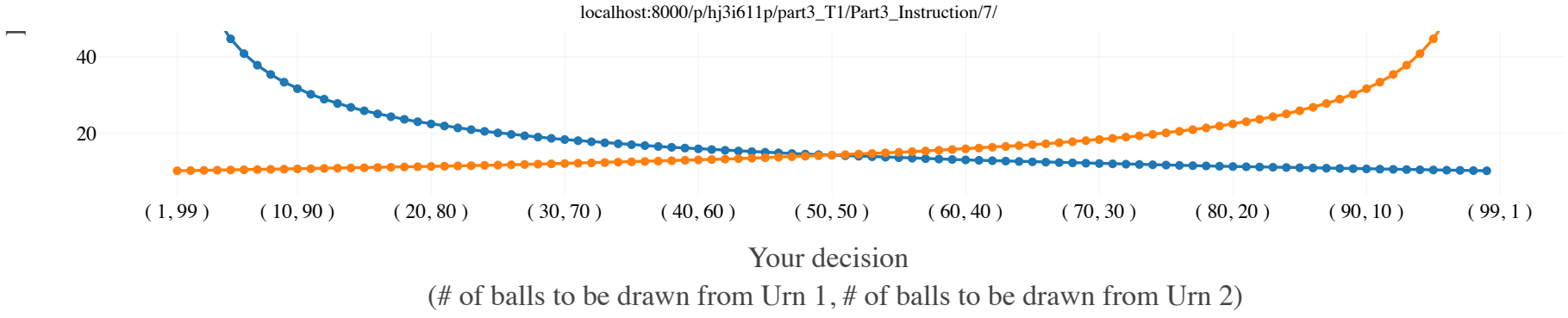
(2) How many balls you decide to draw from each urn:

How many points do you earn for each of the drawn Green balls? This depends on how many balls you decided to draw from the two urns. *The more balls you decided to draw from one urn, every Green ball drawn from that urn awards you the less points while every Green ball drawn from the other urn awards you the more points.* The below figure visualizes the relationship. The x-axis represents your decisions on how many balls to draw from each urn. As the x-value gets closer to the right, it means you draw more balls from Urn 1 and draw less balls from Urn 2. For example, the very left value on the x-axis (1,99) indicates you draw 1 ball from Urn 1 and 99 balls from Urn 2. The very right value on the x-axis (99,1) indicates you draw 99 balls from Urn 1 and 1 ball from Urn 2. The y-axis represents the number of points you earn for every Green ball drawn from each urn. If you take your mouse cursor on the figure, it displays exact values.

Here are some examples of how you read the figure.

- Suppose you decides to draw 10 balls from Urn 1 and 90 balls from Urn 2. The decision is denoted (10,90) on the x-axis. Take your mouse cursor to where the value on the x-axis equals (10,90). Then, you will see the corresponding information that every green ball drawn from Urn 1 awards you 31.62 points (blue line) and that every green ball drawn from Urn 2 awards you 10.54 points (orange line).
- Suppose you decides to draw 50 balls from Urn 1 and 50 balls from Urn 2. The decision is denoted (50,50) on the x-axis. Take your mouse cursor on the center of the figure where the value on the x-axis equals (50,50). Then, you will see the corresponding information that every green ball drawn from Urn 1 awards you 14.14 points (blue line) and that every green ball drawn from Urn 2 also awards you 14.14 points (orange line).
- Suppose you decides to draw 90 balls from Urn 1 and 10 balls from Urn 2. The decision is denoted (90,10) on the x-axis. Take your mouse cursor to where the value on the x-axis equals (90,10). Then, you will see the corresponding information that every green ball drawn from Urn 1 awards you 10.54 points (blue line) and that every green ball drawn from Urn 2 awards you 31.62 points (orange line).





What should you do to make a good decision?

What should you do to earn as many points as possible and increase the probability of winning the prize of 200 Experimental Dollars? On one hand, you should draw more balls from the urn that, you believe, contains more Green balls because you will earn more points as you get to draw more Green balls. On the other hand, you do not want to draw too many balls from the urn even if you are very confident that it has far more Green balls than the other urn. The more balls are drawn from one urn, you earn the less points for a Green ball drawn from that urn.

To help your decision-making, we provide a simulator in every decision-making screen. To use the simulator, you must report your guesses about how many Green balls you and your teammate received in Part 1. Conditional on your guesses, the simulator predicts the expected number of points you will earn for every possible decision. By *"the expected number of points"*, we mean the average number of points you will earn if you repeatedly make the same decision infinitely many times. The simulator can inform you the optimal decision that, conditional on your reported guesses, maximizes the expected number of points you will earn and hence guarantees the highest probability of winning the prize.

Practice decision-making!

Here, you can play around with the simulator and practice your decision-making. The below is the decision-making screen you will face in the actual decision-making rounds.

There is the simulator on the top left of the decision-making screen. Note that your guesses about the number of Green balls you and your teammate received are set to be zero by default. To revise your guesses, you should enter an integer from 0-100 for each of the guesses (the minimum and the maximum numbers of Green balls one can receive are 0 and 100). Once you click "Calculate" button, you will see the corresponding simulation result. You may enter different guesses as many times as you wish. Whenever you enter new guesses, you must click "Calculate" button so that the simulator updates the simulation results accordingly.

How you interpret the simulation results? First, the x-axis represents your potential decisions. As the x-value gets closer to the right, it means more balls are drawn from Urn 1 and less balls are drawn from Urn 2. For example, the very left value on the x-axis (1,99) indicates you draw 1 ball from Urn 1 and 99 balls from Urn 2. The very right value on the x-axis (99,1) indicates you draw 99 balls from Urn 1 and 1 ball from Urn 2. Second, the y-axis represents the expected number of points you will earn, conditional on your entered guesses. Taking your mouse cursor on the simulation results displays the exact values.

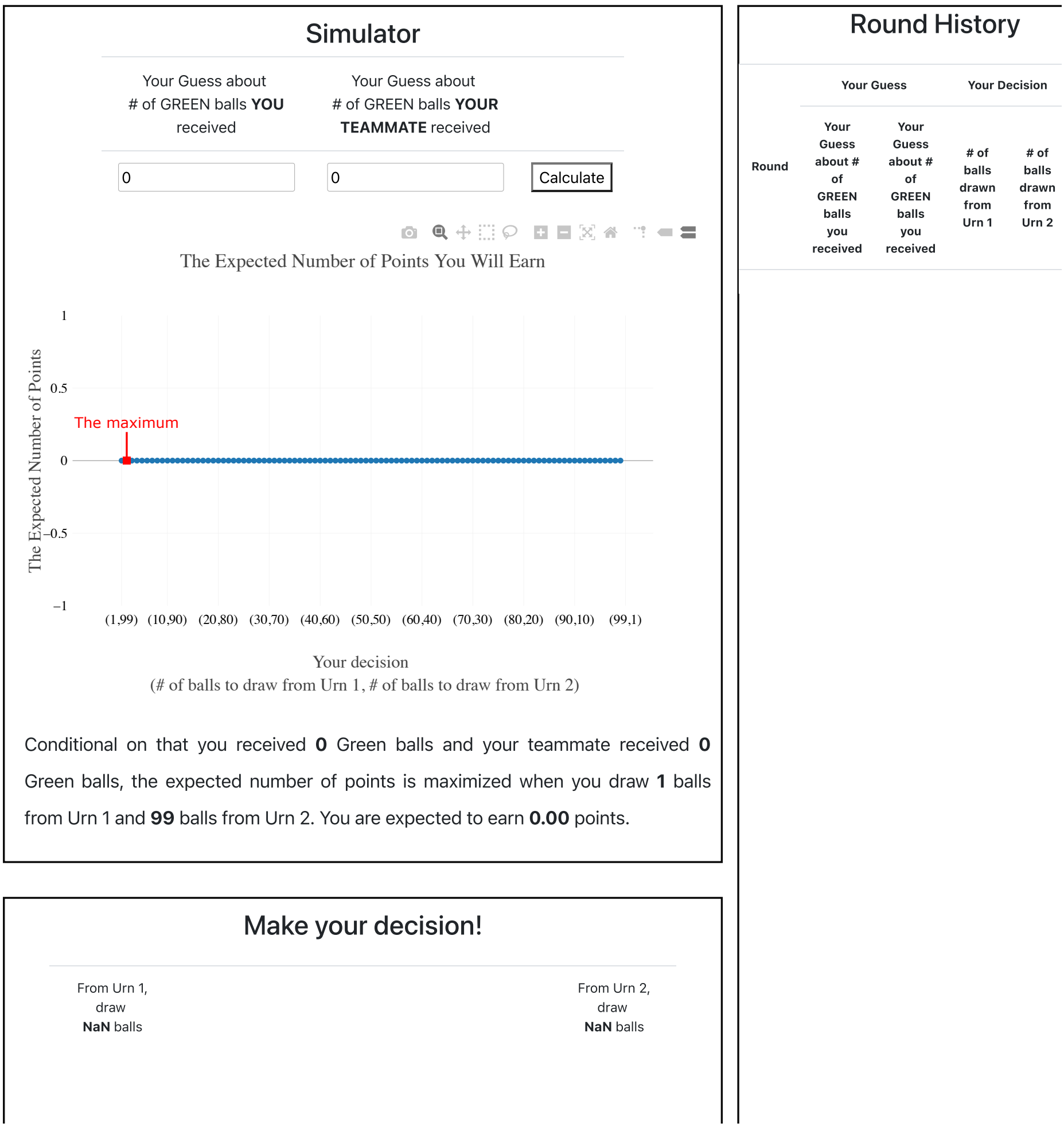
As an example, let's suppose you are guessing you received 40 Green balls and your teammate received 60 Green balls.

- Using the below practice simulator, enter 40 and 60 for your guesses about yourself and about your teammate, respectively. Then click "Calculate" button.
- Once you have done this, you will see the simulator update the result. The simulation result (i.e., the blue dots) shows every pair of your potential decision and the expected number of points you will earn, conditional on the entered guesses that you received 40 Green balls and your teammate received 60 Green balls.

- For instance, conditional on the entered guesses, you are expected to earn 6.37 points when you draw 1 ball from Urn 1 and 99 balls from Urn 2 (the very left of the x-axis). Conditional on the entered guesses, you are expected to earn 4.58 points when you draw 99 balls from Urn 1 and 1 ball from Urn 2 (the very right of the x-axis). Also, notice that, the optimal decision that, conditional on your entered guesses, maximizes the expected number of points you will earn is tagged in red. In this example, conditional on the entered guesses that you received 40 Green balls and your teammate received 60 Green balls, you are expected to earn 7.21 points, which is the maximum, when you draw 31 balls from Urn 1 and 69 balls from Urn 2. The simulator also announces the optimal decision at the bottom of the simulation results.

To indicate your decision, you should use the decision-slider located at the bottom left of the decision-making screen. You can submit your decision by clicking "Submit" button, which will appear at the bottom right of the screen in 20 seconds after each round begins. Once you have submitted your decision, you may not reverse your decision. Afterward, you will receive information about the number of points you actually earned in that round.

To help you keep track of your decisions over time, we also provide you with a history box located on the right of the decision-making screen.



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Debug info

vars_for_template

max_lt	25
num_game_round	50
prize	200

Basic info

ID in group	2
Group	674
Round number	1
Participant	P2
Participant label	
Session code	6hblo8hf